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Inventor(s)	Lu; Ho-Lung et al.

Mobile phone control handle assembly

Abstract

A mobile phone control handle assembly is suitable for holding a mobile phone, and is connected to the mobile phone. The mobile phone control handle assembly includes a handle body, a control module, two handle auxiliary accessories, and at least two magnetic shielding members. The handle body is used to hold the mobile phone. The handle body includes buttons and joysticks. When at least one of the buttons and the joysticks is pressed, the control module transmits a corresponding control signal to the mobile phone. The two handle auxiliary accessories are respectively and detachably fixed on two gripping parts of the handle body through magnets. The two magnetic shielding members are used to shield magnetic fields of the magnets. The two handle auxiliary accessories are configured to enhance a gripping experience of a user when gripping the two gripping parts.

Inventors: **Lu; Ho-Lung (Taipei, TW), Tseng; Tzu-Hua (New Taipei, TW)**

Applicant: **DEXIN CORPORATION** (New Taipei, TW)

Family ID: **90053448**

Assignee: **DEXIN CORPORATION (New Taipei, TW)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4459578	12/1983	Sava	345/161	G05G 9/047
6530838	12/2002	Ha	345/169	A63F 13/24
7580728	12/2008	Vance	455/575.1	A63F 13/22
7653771	12/2009	Liberty	463/37	G06F 1/1632
7733637	12/2009	Lam	361/679.08	G06F 3/0202
7833097	12/2009	Maddox	463/47	A63F 13/23
8417296	12/2012	Caballero	455/566	H04W 52/367
8529357	12/2012	Joynes	463/47	G06F 3/0219
8547340	12/2012	Sizelove	345/173	A63F 13/98
8944912	12/2014	Joynes	463/36	G06F 1/1632
8972617	12/2014	Hirschman	463/37	G06F 3/023
9529447	12/2015	Hodges	N/A	A63F 13/24
9808713	12/2016	Townley	N/A	A63F 13/98
10194509	12/2018	Jorgensen	N/A	H05B 47/1965
10335676	12/2018	Gohara	N/A	A63F 13/24
10471346	12/2018	Yamashita	N/A	A63F 13/56
10596459	12/2019	Kurihara	N/A	A63F 13/25
10661163	12/2019	Gohara	N/A	A63F 13/57
10816976	12/2019	Kamalakantha	N/A	G08C 17/00
10868436	12/2019	Chen	N/A	H01M 50/296
11167209	12/2020	Lu	N/A	A63F 13/98
D945531	12/2021	Wen	D21/333	N/A
D973774	12/2021	Day	D21/333	N/A
11565173	12/2022	Black	N/A	G06F 3/023
11628352	12/2022	Black	463/37	A63F 13/24
11640221	12/2022	Kakinoki	345/174	G06F 3/04166

11707670	12/2022	Khaira	463/37	A63F 13/214
11794099	12/2022	Kamentser	N/A	A63F 13/92
11826642	12/2022	Khaira	N/A	A63F 13/23
11839810	12/2022	Khaira	N/A	A63F 13/24
11911694	12/2023	Townley	N/A	A63F 13/92
11934718	12/2023	Harada	N/A	G06F 3/1211
12061750	12/2023	Harmon	N/A	B60K 35/00
12070678	12/2023	Maker	N/A	A63F 13/92
12074946	12/2023	Wei	N/A	A63F 13/48
12115443	12/2023	Khaira	N/A	A63F 13/92
12121800	12/2023	Maker	N/A	A63F 13/285
12263400	12/2024	Lake	N/A	A63F 13/92
12268956	12/2024	Khaira	N/A	A63F 13/25
12303775	12/2024	Yeo	N/A	A63F 13/98
12324983	12/2024	O'Connor	N/A	A63F 13/92
12346505	12/2024	Nishizawa	N/A	G06F 3/04883
2005/0107165	12/2004	Mendelson	463/6	A63F 13/22
2005/0225530	12/2004	Evans	345/156	A63F 13/22
2006/0252537	12/2005	Wu	463/36	A63F 13/92
2009/0033522	12/2008	Skillman	341/23	G06F 1/1692
2011/0084657	12/2010	Toya	320/108	H02J 7/0044
2011/0199325	12/2010	Payne	345/173	G06F 3/041
2011/0230178	12/2010	Jones	455/566	H04M 1/0241
2011/0234154	12/2010	Navid	320/108	H02J 7/0045
2011/0260969	12/2010	Workman	345/161	G06F 3/0202
2013/0005401	12/2012	Rosenhan	455/557	H04M 1/724092
2013/0052956	12/2012	McKell	361/679.41	H04M 1/05
2013/0095925	12/2012	Xu	463/37	G06F 1/206
2013/0184077	12/2012	Galpern	463/37	A63F 13/24
2013/0267322	12/2012	South	463/37	A63F 13/24
2014/0094309	12/2013	Osawa	463/37	G06F 3/038
2014/0206451	12/2013	Helmes	463/39	A63F 13/23
2014/0247246	12/2013	Maus	345/174	G06F 3/0393
2014/0302921	12/2013	Smith	463/31	A63F 13/23
2015/0084900	12/2014	Hodges	345/173	A63F 13/2145
2016/0361641	12/2015	Koizumi	N/A	A63F 13/2145
2019/0197272	12/2018	Paolizzi	N/A	G06K 7/10881

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3 903 895	12/2018	EP	A63F 13/22
WO 2014/123828	12/2013	WO	G06F 1/16
WO 2021/163557	12/2020	WO	G06F 3/045

OTHER PUBLICATIONS

Machine translation of publication CN 103581392 A, application CN 2013-10185637 A, obtained from STIC, Jul. 1, 2025. cited by examiner
Machine translation of publication, JP 3241602 U, application JP 2023-000454 U, obtained from STIC, Jul. 1, 2025. cited by examiner
Machine translation of publication JP 2023-514961 A, application JP 2022-543580, obtained from STIC, Jul. 1, 2025. cited by examiner
Machine translation of publication CN 206980054 U, application CN 2017-20517196 U, obtained from STIC, Jul. 1, 2025. cited by examiner
Machine translation of publication JP 2009-146737 A, application JP 2007-322964, obtained from STIC, Jul. 1, 2025. cited by examiner
Machine translation of publication TW 1599389, application TW 105101455 A, obtained from STIC, Jul. 1, 2025. cited by examiner
Machine translation of publicaiton JP 2022-517614 A, application JP 2021-540522, obtained from STIC, Jul. 1, 2025. cited by examiner
Machine translation of publication JP 2016-511885 A, application JP 2015-556215, obtained from STIC, Jul. 1, 2025. cited by examiner

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Primary Examiner: Lewis; David L

Assistant Examiner: Hoel; Matthew D

Attorney, Agent or Firm: Li & Cai Intellectual Property Office

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION
(1) This application claims the benefit of priority to Taiwan Patent Application No. 112118604, filed on May 19, 2023. The entire content of the above identified application is incorporated herein by reference.
(2) Some references, which may include patents, patent applications and various publications, may be cited and discussed in the description of this disclosure. The citation and/or discussion of such references is provided merely to clarify the description of the present disclosure and is not an admission that any such reference is “prior art” to the disclosure described herein. All references cited and discussed in this specification are incorporated herein by reference in their entireties and to the same extent as if each reference was individually incorporated by reference.
FIELD OF THE DISCLOSURE
(3) The present disclosure relates to a control handle assembly, and more particularly to a mobile phone control handle assembly.
BACKGROUND OF THE DISCLOSURE
(4) For convenience of carrying, a conventional game handle for being mounted on a mobile phone has a relatively small overall volume. Hence, when gripping the game handle, a user may have a poor gripping experience and a poor operating experience.
SUMMARY OF THE DISCLOSURE
(5) In response to the above-referenced technical inadequacy, the present disclosure provides a mobile phone control handle assembly to improve a poor gripping experience and a poor operating experience of a conventional game handle.

(6) In order to solve the above-mentioned problem, one of the technical aspects adopted by the present disclosure is to provide a mobile phone control handle assembly, which is suitable for holding a mobile phone and being connected to the mobile phone. The mobile phone control handle assembly includes a handle body, a control module, two handle auxiliary accessories, and at least two magnetic shielding members. The handle body includes a connection mechanism and two gripping parts. The connection mechanism and the two gripping parts are configured to jointly hold the mobile phone. The two gripping parts are connected to the connecting mechanism, and the two gripping parts are respectively located on two opposite sides of the connection mechanism. Each of the gripping parts includes a plurality of operating elements disposed on a side surface thereof, and each of the gripping parts includes at least one first magnetic attraction member disposed on another side surface thereof. The control module is disposed in the handle body. The control module is configured to connect to the mobile phone, and the control module is configured to transmit a control signal to the mobile phone. The control signal is generated by operating at least one of the operating elements. Each of the handle auxiliary accessories includes an accessory body and at least one second magnetic attraction member. The accessory body of each of the handle auxiliary accessories is capable of being magnetically attracted to the first magnetic attraction member of one of the gripping parts through the second magnetic attraction member thereof, so as to be fixedly arranged on the another side surface of the one of the gripping parts. The two handle auxiliary accessories are configured to improve a gripping experience for a user on gripping the two gripping parts. A position adjacent to the second magnetic attraction member of each of the handle auxiliary accessories and the first magnetic attraction member of a corresponding one of the gripping parts is provided with at least one of the magnetic shielding members, and each of the magnetic shielding members is configured to reduce an influence of magnetic fields of the first magnetic attraction member and the second magnetic attraction member adjacent thereto on the mobile phone.
(7) Therefore, in the mobile phone control handle assembly provided by the present disclosure, by virtue of “the two handle auxiliary accessories”, the user can have an improved gripping experience and an improved operating experience after mounting the mobile phone on the handle body and fixing the two handle auxiliary accessories on the two gripping parts of the handle body. In this way, the problem of poor gripping experience in the conventional game handle that is mounted on the mobile phone can be effectively solved. In addition, by virtue of “the magnetic shielding members”, an antenna or a magnetic sensitive element of the mobile phone can be effectively prevented from being affected by the magnetic field of the first magnetic attraction member or the second magnetic attraction member. After the mobile phone is mounted on the mobile phone control handle assembly, the antenna or related magnetic sensitive elements of the mobile phone are not affected by the first magnetic attraction member or the second magnetic attraction member.
(8) These and other aspects of the present disclosure will become apparent from the following description of the embodiment taken in conjunction with the following drawings and their captions, although variations and modifications therein may be affected without departing from the spirit and scope of the novel concepts of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The described embodiments may be better understood by reference to the following description and the accompanying drawings, in which:
(2) FIGS. **1** to **3** are schematic exploded views of a mobile phone control handle assembly according to an embodiment of the present disclosure from different viewing angles;
(3) FIGS. **4** and **5** are schematic assembled views of the mobile phone control handle assembly

according to the embodiment of the present disclosure from different viewing angles;
(4) FIG. **6** is a schematic side view of the mobile phone control handle assembly according to the embodiment of the present disclosure; and
(5) FIG. **7** and FIG. **8** are schematic exploded views of the mobile phone control handle assembly according to another embodiment of the present disclosure from different viewing angles.
DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS
(6) The present disclosure is more particularly described in the following examples that are intended as illustrative only since numerous modifications and variations therein will be apparent to those skilled in the art. Like numbers in the drawings indicate like components throughout the views. As used in the description herein and throughout the claims that follow, unless the context clearly dictates otherwise, the meaning of "a", "an", and "the" includes plural reference, and the meaning of "in" includes "in" and "on". Titles or subtitles can be used herein for the convenience of a reader, which shall have no influence on the scope of the present disclosure.
(7) The terms used herein generally have their ordinary meanings in the art. In the case of conflict, the present document, including any definitions given herein, will prevail. The same thing can be expressed in more than one way. Alternative language and synonyms can be used for any term(s) discussed herein, and no special significance is to be placed upon whether a term is elaborated or discussed herein. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms is illustrative only, and in no way limits the scope and meaning of the present disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given herein. Numbering terms such as "first", "second" or "third" can be used to describe various components, signals or the like, which are for distinguishing one component/signal from another one only, and are not intended to, nor should be construed to impose any substantive limitations on the components, signals or the like.
(8) Reference is made to FIGS. **1** to **6**. FIGS. **1** to **3** are respectively disassembled schematic views of different viewing angles of a mobile phone control handle assembly according to an embodiment of the present disclosure. FIGS. **4** and **5** are respectively assembled schematic views of different viewing angles of the mobile phone control handle assembly according to the embodiment of the present disclosure. FIG. **6** is a schematic side view of the mobile phone control handle assembly according to the embodiment of the present disclosure.
(9) A mobile phone control handle assembly **100** of the embodiment of the present disclosure is suitable for holding a mobile phone **200**, and is electrically and/or signally connected to the mobile phone **200**. The mobile phone control handle assembly **100** includes a handle body **1**, a control module **2**, two handle auxiliary accessories **3**, and two magnetic shielding members **4**. The handle body **1** includes a connection mechanism **11** and two gripping parts **12**. The two gripping parts **12** are connected to each other through the connection mechanism **11**. An external shape and an internal structure of the connection mechanism **11** are not specifically limited in the present disclosure.
(10) In a specific embodiment of the present disclosure, the connection mechanism **11** includes two connection components **111** and a pivot component **112** pivotally connected between the two connection components **111**. The two gripping parts **12** are connected to each other through two ends of the two connection components **111** and the pivot component **112**. Through the pivot component **112**, one of the two gripping parts **12** and one of the two connection components **111** that is connected thereto can be operated by a user to rotate relative to another one of the two gripping parts **12** and another one of the two connection components **111**. The handle body **1** further includes a connector **13** that is disposed on one of the two gripping parts **12** (i.e., the gripping part **12** at a right side of FIG. **1**). The connector **13** can be used to connect to a connection hole of the mobile phone **200**. For example, the connector **13** can be a Universal Serial Bus Type-C (USB-C), a micro-USB connector, or a Lightning connector from Apple, Inc., but the present

disclosure is not limited thereto.
(11) It should be noted that, in another embodiment of the present disclosure, the connection mechanism **11** does not include the pivot component **112**. Alternatively, the connection mechanism **11** includes components such as rails and/or sliders. The two gripping parts **12** can move towards or away from each other through the rails and/or the sliders. After the two gripping parts **12** are operated to move towards each other through the rails and/or the sliders, an overall width of the handle body **1** can be reduced. Accordingly, the mobile phone control handle assembly **100** can be convenient for being carried around by the user.
(12) In still another embodiment of the present disclosure, the handle body **1** further includes a spring member. When one of the two gripping parts **12** is pulled away from another one of the two gripping parts **12**, the spring member disposed in the handle body **1** generates an elastic restoring force that enables the two gripping parts **12** to firmly clamp the mobile phone **200**.
(13) Each of the gripping parts **12** further includes a plurality of operating elements. The operating elements can include, for example, a plurality of buttons **121** and at least one joystick **122**. The control module **2** is disposed in the handle body **1**. The control module **2** can include a circuit board, a microprocessor, and a communication chip. When any one of the buttons **121** and the joysticks **122** that are disposed on the gripping parts **12** is operated, the control module **2** can correspondingly generate and transmit a control signal to the mobile phone **200** through the connector **13** or a wireless communication member, so as to operate the mobile phone **200**. For example, after the connection hole of the mobile phone **200** is connected to the connector **13** of the handle body **1** by the user, the user can grip the two gripping parts **12** by both hands, and play games installed in the mobile phone **200** by operating the buttons **121** and the joysticks **122** disposed on the two gripping parts **12**. In a different embodiment, one of the operating elements included in the gripping parts **12** can be a touch pad. When the user touches the touch pad, the control module **2** can correspondingly generate and transmit another control signal to the mobile phone **200**.
(14) As shown in FIG. **3**, two ends of each of the gripping parts **12** are respectively defined as a front end **123** and a rear end **124**. A thickness variation of each of the gripping parts **12** is not greater than 20% from the front end **123** to the rear end **124**. For example, if a minimum thickness D**1** of the front end **123** of each of the gripping parts **12** is 5 centimeters (cm), a maximum thickness D**2** of the rear end **124** thereof ranges from 4 centimeters to 6 centimeters. In other words, the thickness variation from the front end **123** to the rear end **124** of each of the gripping parts **12** is relatively small, which makes the handle body **1** easy to carry. However, during actual use, the user may have a poor gripping experience. Furthermore, when the user operates the buttons **121** and the joysticks **122**, the user may have a poor operating experience.
(15) It is worth mentioning that different users may have different palm sizes. When a user having a small palm size grips the handle body **1**, the user may not have the poor gripping experience. However, when a user having a large palm size grips the handle body **1**, the user may have the poor gripping experience. When gripping the handle body **1**, the user with the poor gripping experience can further mount the two handle auxiliary accessories **3** respectively on rear sides **125** of the two gripping parts **12**. Accordingly, the two handle auxiliary accessories **3** can respectively increase thicknesses of the rear ends **124** of the two gripping parts **12**, so as to enhance the gripping experience and the operating experience of the user during gripping and operation of the mobile phone control handle assembly **100**.
(16) More specifically, the rear side **125** of each of the gripping parts **12** has two first magnetic attraction members **126** and two first engagement structures **127** (as shown in FIG. **1** to FIG. **3**). Each of the first engagement structures **127** can be, for example, a groove. Each of the first magnetic attraction members **126** can be, for example, a magnet or a structure capable of being attracted by the magnet. Each of the first magnetic attraction members **126** is correspondingly disposed in one of the grooves (i.e., the first engagement structures **127**). In each of the gripping

parts **12**, the quantity and location of the first magnetic attraction members **126**, the quantity and location of the first engagement structures **127**, and the appearance of the first engagement structures **127** are not limited to those shown in the drawings.

(17) Each of the handle auxiliary accessories **3** includes an accessory body **31**, two second magnetic attraction members **32**, two second engagement structures **33**, and a magnetic shielding member **4**. In each of the handle auxiliary accessories **3**, the quantities and locations of the second magnetic attraction members **32**, the second engagement structures **33**, and the magnetic shielding member **4** can be changed according to practical requirements, and are not limited to those shown in the drawings.

(18) Two ends of the accessory body **31** are respectively defined as a front end **31**A and a rear end **31**B. A thickness of partial sections of the accessory body **31** gradually increases from the front end **31**A to the rear end **31**B. When the accessory body **31** of one of the handle auxiliary accessories **3** is fixed on the rear side **125** of one of the gripping parts **12**, the rear end **31**B of the accessory body **31** is located at the rear end **124** of the gripping part **12**. For example, the accessory body **31** can have a minimum thickness D**3** at the front end **31**A, and the accessory body **31** can have a maximum thickness D**4** at the rear end **31**B (as shown in FIG. **6**). The thickness of the accessory body **31** is gradually increased from the position where the accessory body **31** has the minimum thickness D**3** to the position where the accessory body **31** has the maximum thickness D**4**.

(19) Each of the second magnetic attraction members **32** and a corresponding one of the first magnetic attraction members **126** are members capable of magnetically attracting each other. For example, each of the second magnetic attraction members **32** can be a strong magnet, and a corresponding one of the first magnetic attraction members **126** can be a member capable of being attracted by the strong magnet. Conversely, the first magnetic attraction members **126** can be a strong magnet, and the second magnetic attraction members **32** can be a member capable of being attracted by the strong magnet.

(20) Each of the second engagement structures **33** can be engaged with a corresponding one of the first engagement structures **127**. The two second magnetic attraction members **32** and the two second engagement structures **33** disposed on each of the accessory bodies **31** can be respectively connected to the two first magnetic attraction members **126** and the two first engagement structures **127** of a corresponding one of the gripping parts **12**. Accordingly, the accessory body **31** can be fixed in position on the rear side **125** of the gripping part **12**. For example, an appearance of each of the second engagement structures **33** can be a hollow cylindrical structure, and each of the second magnetic attraction members **32** can be correspondingly disposed in the hollow cylindrical structure. As such, when each of the hollow cylindrical structures (i.e., the second engagement structure **33**) is engaged with a corresponding groove (i.e., the first engagement structure **127**), the second magnetic attraction member **32** is magnetically attracted to the first magnetic attraction member **126**.

(21) According to the above configuration, after the accessory body **31** is fixed to the rear side **125** of the corresponding gripping part **12** through the two second magnetic attraction members **32** and the two second engagement structures **33**, the rear end **31**B of the accessory body **31** is adjacent to the rear end **124** of the corresponding gripping part **12**. Therefore, when the user holds the mobile phone control handle assembly **100**, the user can grip the two accessory bodies **31** and the two handle parts **12** at the same time, and thus have a relatively better gripping experience (especially for the user with a large palm size).

(22) Reference is made to FIG. **3** and FIG. **6**. More specifically, when the user having a large palm size holds the handle body **1** (as shown in FIG. **3**), the user may have a relatively poor gripping experience due to the thickness D**2** of the handle body **1** at the rear end **124** being relatively small. At this time, the user can mount the handle auxiliary accessory **3** on the rear side **125** of the gripping part **12**. When the user holds the mobile phone control handle assembly **100** (as shown in FIG. **6**), the user may have a relatively better gripping experience due to the thickness D**5** of the

mobile phone control handle assembly **100** at the rear end **124** being obviously greater than the thickness D**2** as shown in FIG. **3**.

(23) Each of the handle auxiliary accessories **3** can include at least one magnetic shielding member **4** disposed adjacent to the second magnetic attraction members **32** and the first magnetic attraction members **126** that corresponds in position to the second magnetic attraction members **32**. The magnetic shielding member **4** is configured to reduce an influence of magnetic fields of the adjacent first magnetic attraction members **126** and the adjacent second magnetic attraction members **32** on the mobile phone **200**. In a practical application, the magnetic shielding member **4** can be, for example, a magnetic shielding plate or a magnetic shielding block. The magnetic shielding member **4** can be disposed at a position adjacent to a member that is a magnet. For example, if the first magnetic attraction member **126** is the magnet, the magnetic shielding member **4** is disposed at a position adjacent to the first magnetic attraction member **126** on the gripping part **12**. Conversely, if the second magnetic attraction member **32** is the magnet, the magnetic shielding member **4** is disposed at a position adjacent to the second magnetic attraction member **32** on the accessory body **31**.

(24) It should be noted that an upper end (i.e., an earpiece end) and a lower end of a typical smart phone usually have multiple sets of antennas and/or related magnetic sensitive elements. Therefore, when the smart phone is placed on the mobile phone control handle assembly **100** of the present disclosure, the antennas and/or the magnetic sensitive elements of the smart phone are close to the first magnetic attraction members **126** and the second magnetic attraction member **32**. Hence, if the magnetic shielding member **4** is not disposed adjacent to the first magnetic attraction members **126** or the second magnetic attraction members **32**, the magnetic fields of the magnets (i.e., the first magnetic attraction members **126** or the second magnetic attraction members **32**) may cause interference to the antennas and/or the magnetic sensitive elements of the smart phone, thereby resulting in partial or total malfunction of the smart phone.

(25) Based on the above, through the configuration of the two handle auxiliary accessories, the first magnetic attraction members, the second magnetic attraction members, and the magnetic shielding members, the mobile phone control handle assembly of the present disclosure can improve the gripping experience and the operating experience of the user (especially for the user with a large palm size) during gripping and operation of the mobile phone control handle assembly. Moreover, the first magnetic attraction members and the second magnetic attraction members can be prevented from interfering with the antennas and/or the magnetic sensitive elements of the smart phone.

(26) Referring to FIG. **7** and FIG. **8**, FIG. **7** and FIG. **8** are respectively disassembled schematic views of different viewing angles of a mobile phone control handle assembly according to another embodiment of the present disclosure. A first difference between the present embodiment and the previous embodiment is that the handle body **1** of the present embodiment further includes at least one first rechargeable battery **14**. Each of the gripping parts **12** includes at least one first charging connection member **128**. The quantity and location of the first rechargeable battery **14** disposed in the handle body **1** are not limited to those shown in the drawings.

(27) Each of the handle auxiliary accessories **3** further includes at least one second rechargeable battery **34** and at least one second charging connection member **35**. When the two handle auxiliary accessories **3** are respectively fixed on the rear sides **125** of the two gripping parts **12**, the second charging connection member **35** of each of the handle auxiliary accessories **3** can be connected to the first charging connection member **128** of a corresponding one of the gripping parts **12**. The control module **2** is configured to enable the second rechargeable battery **34** to charge the first rechargeable battery **14**, or to enable the first rechargeable battery **14** to charge the second rechargeable battery **34**. The specific structures of the first charging connection member **128** and the second charging connection member **35** can be, for example, various electrical connection structures (i.e., various metal contacts).

(28) It should be noted that, in the present embodiment, each of the accessory bodies **31** includes one second rechargeable battery **34** disposed therein and two second charging connection members **35** disposed thereon, but the present disclosure is not limited thereto. In a different embodiment, out of the two accessory bodies **31**, only one of the accessory bodies **31** includes the second rechargeable battery **34** and the second charging connection members **35**, and another one of the accessory bodies **31** does not include the second rechargeable battery **34** and the second charging connection members **35**. Similarly, in the present embodiment, each of the gripping parts **12** includes two first charging connection members **128**, but the present disclosure is not limited thereto. In a different embodiment, only one of the gripping parts **12** includes the first charging connection members **128**, and another one of the gripping parts **12** does not include the first charging connection members **128**.
(29) In another embodiment, the control module **2** can decide to enable the first rechargeable battery **14** to charge the second rechargeable battery **34**, or enable the second rechargeable battery **34** to charge the first rechargeable battery **14** according to a first remaining battery power of the first rechargeable battery **14** of the handle body **1** and a second remaining battery power of the second rechargeable battery **34** of each of the handle auxiliary accessories **3**.
(30) Specifically, after the two handle auxiliary accessories **3** are respectively fixed on the rear sides **125** of the two gripping parts **12** by the user, the control module **2** can, for example, firstly obtain the first remaining battery power of the first rechargeable battery **14** and the second remaining battery power of the second rechargeable battery **34**. Then, if the control module **2** determines that the first remaining battery power of the first rechargeable battery **14** is greater than a predetermined percentage (i.e., greater than 80%) and the second remaining battery power of the second rechargeable battery **34** is less than a predetermined percentage (i.e., less than 20%), the control module **2** controls the first rechargeable battery **14** to charge the second rechargeable battery **34**. Conversely, if the control module **2** determines that the second remaining battery power of the second rechargeable battery **34** is greater than the predetermined percentage (i.e., greater than 80%) and the first remaining battery power of the first rechargeable battery **14** is less than the predetermined percentage (i.e., less than 20%), the control module **2** controls the second rechargeable battery **34** to charge the first rechargeable battery **14**. Naturally, how the control module **2** determines a charging method of the first rechargeable battery **14** and the second rechargeable battery **34** according to the first remaining battery power of the first rechargeable battery **14** and the second remaining battery power of the second rechargeable battery **34** can be designed and changed based on actual needs. The above description is merely one illustrative example.
(31) A second difference between the present embodiment and the previous embodiment is that the handle body **1** of the present embodiment further includes a charging port **15**. The charging port **15** is electrically connected to the control module **2**. An external power source can charge the first rechargeable battery **14** through the charging port **15**. The control module **2** can control the second rechargeable battery **34** not to charge the first rechargeable battery **14** when the charging port **15** is connected to the external power source. In a practical application, when the control module **2** determines that the charging port **15** is connected to the external power source, and when the first remaining battery power of the first rechargeable battery **14** is charged to reach a predetermined percentage (i.e., 100%) or exceed another predetermined percentage (i.e., 90%), the control module **2** can control the external power source to further charge the second rechargeable battery **34**. Naturally, in a different embodiment, when the charging port **15** is connected to the external power source, the control module **2** can also control the external power source to charge the first rechargeable battery **14** and the second rechargeable battery **34** at the same time.
(32) A third difference between the present embodiment and the previous embodiment is that each of the handle auxiliary accessories **3** of the present embodiment further includes a processor **36** disposed in the accessory body **31**, and includes a battery indicator **37** and a charging port **38**

arranged on the accessory body **31**. The processor **36** is electrically connected to the second rechargeable battery **34**. The processor **36** can control the battery indicator **37** according to the second remaining battery power of the second rechargeable battery **34**, so that the battery indicator **37** can display the second remaining battery power corresponding to the second rechargeable battery **34**. That is, the user can connect the charging port **38** of each of the accessory bodies **31** with a relevant connecting wire, so that the external power source can charge the second rechargeable battery **34**. By monitoring the battery indicator **37**, the user can find out whether or not the second rechargeable battery **34** has been charged completely. The second rechargeable battery **34** is electrically connected to the charging port **38**. The external power can charge the second rechargeable battery **34** through the charging port **38**.
(33) It is worth mentioning that the battery indicator **37** can display only whether or not the second rechargeable battery **34** is charged completely, or the battery indicator **37** can also display a battery power percentage of the second rechargeable battery **34**, but the present disclosure is not limited thereto.
(34) The multiple differences of the present embodiment from the previous embodiment are not limited to being co-existent. Each difference can also be incorporated into the previous embodiment to form a new embodiment with the different feature.
(35) The foregoing description of the exemplary embodiments of the disclosure has been presented only for the purposes of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Many modifications and variations are possible in light of the above teaching.
(36) The embodiments were chosen and described in order to explain the principles of the disclosure and their practical application so as to enable others skilled in the art to utilize the disclosure and various embodiments and with various modifications as are suited to the particular use contemplated. Alternative embodiments will become apparent to those skilled in the art to which the present disclosure pertains without departing from its spirit and scope.

Claims

1. A mobile phone control handle assembly, which is suitable for holding a mobile phone and is connected to the mobile phone, the mobile phone control handle assembly comprising: a handle body including a connection mechanism and two gripping parts, wherein the connection mechanism and the two gripping parts are configured to jointly hold the mobile phone, the two gripping parts are connected to the connecting mechanism, and the two gripping parts are respectively located on two opposite sides of the connection mechanism; wherein each of the gripping parts includes a plurality of operating elements disposed on a side surface thereof, and each of the gripping parts includes at least one first magnetic attraction member disposed on another side surface thereof; a control module disposed in the handle body, wherein the control module is configured to connect to the mobile phone, the control module is configured to transmit a control signal to the mobile phone, and the control signal is generated by operating at least one of the operating elements; two handle auxiliary accessories, each of the handle auxiliary accessories including an accessory body and at least one second magnetic attraction member, wherein the accessory body of each of the handle auxiliary accessories is capable of being magnetically attracted to the first magnetic attraction member of one of the gripping parts through the second magnetic attraction member, so as to be fixedly arranged on the another side surface of the one of the gripping parts; wherein the two handle auxiliary accessories are configured to enhance a gripping experience of a user when gripping the two gripping parts; and at least two magnetic shielding members, wherein at least one of the magnetic shielding members is disposed at a position adjacent to the second magnetic attraction member of each of the handle auxiliary accessories and the first magnetic attraction member of a corresponding one of the gripping parts,

and each of the magnetic shielding members is configured to reduce an influence of magnetic fields of the adjacent first magnetic attraction member and the adjacent second magnetic attraction member on the mobile phone.
2. The mobile phone control handle assembly according to claim 1, wherein each of the gripping parts has at least one first engagement structure, each of the accessory bodies has at least one second engagement structure, and the second engagement structure of each of the accessory bodies is engageable with the first engagement structure of a corresponding one of the gripping parts, so that each of the accessory bodies is capable of being fixed to a corresponding one of the gripping parts.
3. The mobile phone control handle assembly according to claim 2, wherein, in each of the gripping parts, the first magnetic attraction member is disposed in the first engagement structure; wherein, in each of the handle auxiliary accessories, the second magnetic attraction member is disposed in the second engagement structure; wherein, in each of the gripping parts and a corresponding one of the handle auxiliary accessories, the first magnetic attraction member and the second magnetic attraction member are magnetically attracted to each other when the first engagement structure is engaged with the second engagement structure.
4. The mobile phone control handle assembly according to claim 1, wherein the handle body further includes a first rechargeable battery, each of the gripping parts includes a first charging connection member, and at least one of the handle auxiliary accessories further includes at least one second rechargeable battery and a second charging connection member; wherein, when the second charging connection member is connected to the first charging connection member, electric energy of the second rechargeable battery is provided to at least one of the control module and the first rechargeable battery through the second charging connection member and the first charging connection member.
5. The mobile phone control handle assembly according to claim 4, wherein, when each of the accessory bodies is fixed to one of the gripping parts through the second magnetic attraction member, the second charging connection member is connected to the first charging connection member, and the control module enables electric energy of the first rechargeable battery to be transferred to the second rechargeable battery through the first charging connection member and the second charging connection member, so as to charge the second rechargeable battery.
6. The mobile phone control handle assembly according to claim 5, wherein the control module is electrically connected to the first rechargeable battery, and the control module is electrically connected to the second rechargeable battery through the first charging connection member and the second charging connection member; wherein, according to a first remaining battery power of the first rechargeable battery and a second remaining battery power of the second rechargeable battery, the control module determines whether to enable the first rechargeable battery to charge the second rechargeable battery, or to enable the second rechargeable battery to charge the first rechargeable battery.
7. The mobile phone control handle assembly according to claim 4, wherein the handle body further includes a charging port, the charging port is electrically connected to the control module, an external power source is capable of charging the first rechargeable battery through the charging port, and the control module is configured to control the second rechargeable battery not to charge the first rechargeable battery when the charging port is connected to the external power source.
8. The mobile phone control handle assembly according to claim 4, wherein each of the accessory bodies further includes a charging port, the second rechargeable battery is electrically connected to the charging port, and an external power source is capable of charging the second rechargeable battery through the charging port.
9. The mobile phone control handle assembly according to claim 4, wherein each of the accessory bodies further includes a processor and a battery indicator, the processor is electrically connected to the second rechargeable battery, and the processor is configured to control the battery indicator

according to a remaining battery power of the second rechargeable battery, so that the battery indicator displays the remaining battery power of the second rechargeable battery.

10. The mobile phone control handle assembly according to claim 1, wherein two opposite ends of each of the accessory bodies are respectively defined as a front end and a rear end, and a thickness of partial sections of each of the accessory bodies is gradually increased from the front end to the rear end; wherein, when each of the accessory bodies is fixed on a rear side of one of the gripping parts, the rear end of the accessory body is located at a rear end of the handle body; wherein two opposite ends of each of the gripping parts are respectively defined as a front end and a rear end, and a thickness variation from the front end to the rear end of each of the gripping parts is not greater than 20%.
